

1. (Pascal's Identity) Prove the identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

for integers  $n \geq 1$  and  $1 \leq k \leq n-1$ , using a combinatorial argument.

**Solution:** Count the number of  $k$ -element subsets of an  $n$ -element set, say  $\{1, 2, \dots, n\}$ . There are  $\binom{n}{k}$  such subsets.

Partition these subsets into two disjoint classes: those that do not contain  $n$ , and those that do.

If a subset does not contain  $n$ , then all  $k$  elements must be chosen from  $\{1, \dots, n-1\}$ , giving  $\binom{n-1}{k}$  possibilities.

If a subset does contain  $n$ , then we must choose the remaining  $k-1$  elements from  $\{1, \dots, n-1\}$ , giving  $\binom{n-1}{k-1}$  possibilities.

Summing the two disjoint cases yields

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

2. Suppose we have  $n$  distinct objects and want to distribute them into  $k$  labeled boxes, where box  $i$  contains exactly  $n_i$  objects, with  $n_1 + \dots + n_k = n$ . Show that the number of such distributions is

$$\frac{n!}{n_1!n_2!\dots n_k!}.$$

**Solution:** We count the number of ways to partition  $n$  distinct objects into ordered groups of sizes  $n_1, \dots, n_k$ .

First, arrange all  $n$  objects in a line. There are  $n!$  such permutations.

Now divide this ordering into consecutive blocks: the first  $n_1$  objects go to box 1, the next  $n_2$  to box 2, and so on.

However, this procedure overcounts, since permuting the elements within any given box does not change the assignment. For box  $i$ , there are  $n_i!$  such permutations.

Thus each distribution is counted  $n_1!n_2!\dots n_k!$  times, and the correct count is

$$\frac{n!}{n_1!n_2!\dots n_k!}.$$

3. Give a combinatorial proof of the identity

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \binom{m+n}{k}.$$

**Solution:** Let  $A$  and  $B$  be disjoint sets with  $|A| = m$  and  $|B| = n$ . We count the number of  $k$ -element subsets of  $A \cup B$ .

On one hand, this is simply  $\binom{m+n}{k}$ .

On the other hand, we can count by conditioning on how many elements are chosen from  $A$ . If exactly  $i$  elements are chosen from  $A$ , then  $k - i$  must be chosen from  $B$ . The number of such subsets is

$$\binom{m}{i} \binom{n}{k-i}.$$

Summing over all possible  $i$  gives

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}.$$

Since both expressions count the same quantity, the identity follows.

4. How many solutions in nonnegative integers are there to

$$x_1 + x_2 + x_3 + x_4 = 10?$$

How many solutions are there if we additionally require  $x_1, x_2, x_3, x_4 \geq 1$ ?

**Solution:** We use the stars-and-bars method.

The number of nonnegative integer solutions to

$$x_1 + x_2 + x_3 + x_4 = 10$$

is equal to the number of ways to place 10 identical objects into 4 bins, which is

$$\binom{10 + 4 - 1}{4 - 1} = \binom{13}{3} = 286.$$

If we require  $x_i \geq 1$ , define  $y_i = x_i - 1$ . Then each  $y_i \geq 0$  and

$$y_1 + y_2 + y_3 + y_4 = 6.$$

Thus the number of solutions is

$$\binom{6 + 4 - 1}{4 - 1} = \binom{9}{3} = 84.$$

5. How many distinct permutations are there of the word

MISSISSIPPI?

**Solution:** The word MISSISSIPPI has 11 letters with multiplicities:

$$M^1, I^4, S^4, P^2.$$

If all letters were distinct, there would be  $11!$  permutations. However, permuting identical letters does not produce new words.

There are  $4!$  ways to permute the  $I$ 's among themselves,  $4!$  ways for the  $S$ 's, and  $2!$  ways for the  $P$ 's, all of which leave the word unchanged. Therefore we divide by these factors to correct for overcounting:

$$\frac{11!}{4! 4! 2!}.$$

6. (Hockey-stick identity) Prove that

$$\sum_{i=k}^n \binom{i}{k} = \binom{n+1}{k+1}.$$

**Solution:** We count the number of  $(k+1)$ -element subsets of  $\{1, 2, \dots, n+1\}$ .

Fix such a subset and let  $i+1$  be its largest element. Then the remaining  $k$  elements must be chosen from  $\{1, \dots, i\}$ , which can be done in  $\binom{i}{k}$  ways.

Since the largest element can range from  $k+1$  to  $n+1$ , i.e.,  $i = k, \dots, n$ , summing over all possibilities gives

$$\sum_{i=k}^n \binom{i}{k}.$$

On the other hand, the total number of  $(k+1)$ -element subsets is  $\binom{n+1}{k+1}$ . Equating the two counts proves the identity.